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COURSE CODE : CSA0619

COURSE NAME : DESIGN AND ANALYSIS OF ALGORITHMS

CO2 ASSESSMENT TOOL 1

ANALYTICAL PROBLEM SOLVING

CONVEX HULL - OUTER BOUNDARY OF SURVEY POINTS

Problem Understing :

Give the Survey Points:

$$\{(0,1),(2,4),(5,5),(6,2),(4,0),(2,1)\}$$

The objective is to find the Convex Hall, which is the Smallest Convex Polygon that enclosed all the given Points.

Need For Convex Hull Computation

- Finds the outer boundary of Survey Points
- Used in land surveying and GIS mapping.
- Helps determine the boundary of a Property
- Eliminates interior points from the boundary.

3 **Method Selection :**

Algorithm: Brute force Convex Hull

Steps:

1. Consider every Pair of Points.
2. Draw a line between them
3. Check whether all other Points lie on the Same side.
4. If Yes, the line is a hall edge.

3 Calculations

Number of points = 6

Total line Pairs : $\frac{6(6-1)}{2} = 15$

Comparison

$15 \times 4 = 60$

④ Valid Hull Edges.

- AB
- BC
- CD   Interior Point: F(2,1)
- DE
- EA

⑤ Final Answer

Convex Hull:

A → B → C → D → E → A

⑥ Time Complexity

For n points:

- Line Pairs $= O(n^2)$
- checking remaining points $= O(n)$

Time Complexity $= O(n^3)$

⑦ Interpretation of Results:

The Brute Force Convex Hull algorithm successfully identifies the outer boundary of the survey points. It performs 60 comparisons and has a time complexity of $O(n^3)$. Although simple to implement, it is inefficient for large dataset. In land mapping application, the Convex Hull is useful for determining property

...ndaries, survey regions and geographic mapping